

Lecture 3: Data analysis and visualization (I): Descriptive and exploratory analysis

SOC612: Data Analytics for Social Research

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Today's schedule

- 1 **Administration:** Moodle access for PhDs?
- 2 **Recap from last week:** Some takeaways, Task 1
- 3 **Descriptive and exploratory analysis:** Short input, main steps of describing and exploring data before follow-up inferential analyses
- 4 **Outlook for next week**
- 5 **Tutorial:** *SOC612_Task2.pdf* on Moodle, GitHub

Recap from last week

Some takeaways

- The data we're working with is often messy and in most cases not yet ready for further analysis, so we need to process it.
- This process includes steps for formatting the dataset, recoding and releveling variables, dealing with missing cases, and sometimes more complex variable treatments, like indexing, standardising, or normalising.
- The whole process can be exhausting and time-consuming, but many parts can also be automated using AI assistance.

Recap from last week

Task 1

- Task 1 was more difficult than what a normal data cleaning process would normally look like, and in the future, for all exercises, it won't be so extensive.
- Not everything in all steps is fully specified, so it can be, that in the end, your solution might vary slightly from my solutions.

Descriptive and exploratory analysis

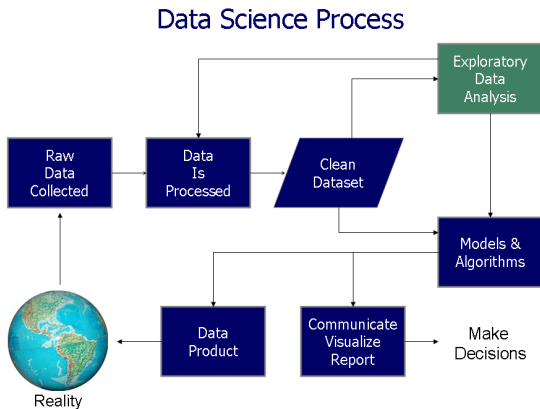
Short input



"You cleaned and prepared the data. Now what?"

Descriptive and exploratory analysis

Short input



https://en.wikipedia.org/wiki/Exploratory_data_analysis#/media/File:Data_visualization_process.v1.png

Descriptive and exploratory analysis

Short input

- Our data is normally too big and abstract to just understand it by looking at it, we need to understand it systematically.
- We need to explore how the data is distributed and what the shape looks like. Samples can have the same point estimates but the distribution can be totally different.
- Leading Questions: What is the shape of my data? How dispersed is it? How many outliers are there?

"Exploratory data analysis is detective work – numerical detective work – or counting detective work – or graphical detective work." (John Tukey, 1977)

Descriptive and exploratory analysis

Short input

Are the following tasks part of the descriptive or exploratory part of a data analysis, or of the inferential or predictive? Why?

In class

- **Q1:** Predict a student's exam score based on their study hours and attendance using a regression model.
- **Q2:** Visualize the relationship between 'age' and 'income' to identify trends or patterns.
- **Q3:** Describe the distribution of a variable like 'sleeping hours' by calculating its mean, median, and standard deviation.

Descriptive and exploratory analysis

Short input

Are the following tasks part of the descriptive or exploratory part of a data analysis, or of the inferential or predictive? Why?

In class

- **Q4:** Test whether there is a statistically significant difference in average income between two education groups.
- **Q5:** Use a hypothesis test to determine if a new marketing strategy significantly increases sales compared to the old one.
- **Q6:** Summarize the frequency of each category in a variable like 'education level' using a bar chart.

Descriptive and exploratory analysis

- We'll go through the steps for describing and exploring a cross-sectional survey dataset.
- We're looking at some lifestyle-related behaviours in the US, using the *National Health Interview Survey-Wave 2022*.
- The script covers the important steps before an inferential analysis, including descriptives and visualizations.
- → RMarkdown. On the LU moodle and GitHub, you'll also find a *.ipynb* file.

Outlook for next week

- Please finish the task from today's tutorial.
- Next week, we'll start with inferential analyses and interpretation and presentation of the results.
- If you have time, please look at the recommended reading.



- Please download *SOC612_Task2.pdf* and the dataset from the SOC612 course on moodle.
- A solution will be uploaded tomorrow morning on moodle and GitHub.

Thanks for coming!

